A Scrum-MOSA (Modular Open Systems Approach) Methodology for Iterative and Incremental System Development

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The file contains the results in narrative form for the research paper, A Scrum-MOSA (Modular Open Systems Approach) Methodology for Iterative and Incremental System Development. The 24 DoD PART questions were evaluated against the proposed sMOSA framework. The file contains narrative results for the research paper, A Scrum-MOSA (Modular Open Systems Approach) Methodology for Iterative and Incremental System Development. The sMOSA methodology is a proposed iterative, incremental agile development approach for delivering a modular, MOSA-compliant system. Traditionally, the Department of Defense (DoD) Program Assessment and Rating Tool (PART) evaluates a program's compliance with MOSA principles. The PART includes 11 Business and Programmatic questions and 13 Technical questions aligned with the five MOSA principles. The proposed sMOSA framework is validated against the PART questions. The results are contained herein.

I. Appendix A: Results and Data Analysis

A. Results and Data Analysis

1. Business-Programmatic Questions

A1: To what extent is MOSA incorporated into the program's acquisition planning? This PART question evaluates whether the MOSA principles were a part of the acquisition strategy, planning activities, and lifecycle processes from the beginning of the program. The program incorporates MOSA objectives, such as affordability, integration, reuse, commonality of components, and interoperability, into acquisition planning activities during the Product Roadmap Planning stage [1]. This effort includes project, product, and scope planning, as documented in the Systems Engineering Plan (SEP) and Conceptual and Contractual Design (CCD). Building the product backlog also involves planning for MOSA compliance, planning releases, the MOSA strategy, IP planning, and an architecture overview to understand how the MOSA principles will be applied. Acquisition and MOSA planning are incorporated into Stage two through the development of the sprint backlog and the selection of user stories that incorporate applicable MOSA principles. The Agile Scrum practice strengthens acquisition planning through stakeholder involvement, collaboration, and iterative planning [2]. For instance, the Product Backlog and Release Plan guide stakeholder acquisition needs, while the MOSA component ensures architectural decisions align with the objectives of adaptability and modularity.

A2: To what extent did the program plan for its implementation of MOSA? This PART question evaluates the program that intentionally and comprehensively plans to execute the MOSA attributes throughout the lifecycle. A1 is related to acquisition planning, and A2 asks about the existence and quality of the MOSA implementation plan. The program establishes specific, measurable, achievable, relevant, and time-phased (SMART) metrics to evaluate how well the MOSA objectives are being met [1]. The MOSA implementation planning and metrics are captured in the MOSA Strategy and verification and validation (V&V) planning. The product scope and Release Plan establish the program's vision and end state, while architectural development provides the ground rules for modularity, interoperability, and open interfaces. The Agile Scrum practices align with the sMOSA Strategy Plan through continuous stakeholder engagement and iterative changes to the Plan as requirements and lessons learned evolve [2].

A3: To what extent is the program's MOSA implementation based on systems engineering principles and processes? This PART question evaluates the program's incorporation of systems engineering lifecycle processes, principles, and methods [1, 3, 4]. The sMOSA methodology incorporates MOSA into the traditional systems engineering activities, such as architectural development, modular decomposition, and interface identification during the sMOSA

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Product Roadmap Planning stages. The SE processes continue in the Development and Integration stage through architectural refinement, interface allocation, continuous verification, and alternative configurations. Agile Scrum supports the IID practice of delivering incremental products and prototypes through iterative cycles and continuous stakeholder feedback. The sMOSA Review and Retrospective stage integrates systems engineering principles through validation, integration testing, compatibility assessments, and verification of architectural objectives.

A4: To what extent are responsibilities assigned for implementing MOSA? This PART question evaluates whether roles and responsibilities were assigned for implementing MOSA throughout the system lifecycle. Successful MOSA implementation with the sMOSA methodology requires coordination with program management, systems engineering, architecture teams, contractors, suppliers, and stakeholders. The sMOSA Product Roadmap Planning stage specifically includes the activity that defines roles and responsibilities. Agile Scrum assigns roles for specific purposes, such as Product Owner, Scrum Master, Development Team, and stakeholders [2]. The Agile Scrum roles expand the sMOSA project's responsibilities to include oversight of modular architecture, interface management, standards governance, and conformance planning.

A5: To what extent is the program staff trained on, or have relevant experience in MOSA concepts and implementation? This PART question evaluates whether personnel have the knowledge, skills, and training necessary to implement the MOSA principles throughout the lifecycle. Training and continuous improvement plans are initially captured during the sMOSA Product Roadmap Planning stage. Agile Scrum supports this through continuous learning and improvement. Personnel are continuously trained, and lessons learned are captured during project reviews and retrospectives. Lessons learned and information debt are captured throughout the sMOSA stages, providing iterative opportunities for training and identifying knowledge gaps.

A6: To what extent does the program's configuration management process encompass changes to key interfaces and corresponding standards? This PART question evaluates whether the program's configuration management processes control the changes to designated key interfaces and related standards. The Agile Scrum process and principles promote welcoming changing requirements and late development changes. MOSA requires that interface changes remain controlled through documented baselines and version-management practices. Although the sMOSA process does not explicitly define a configuration management process or interface change control board, the Project, MOSA, and IP plans provide a framework for managing changes to key interfaces and standards. Iterative development, backlog refinement, architectural development, and sprint reviews allow teams to evaluate and incorporate changes while maintaining traceability. The sMOSA Review and Retrospective stages also test for compatibility and review documented changes.

A7: To what extent have program requirements been analyzed, and refined as needed, to ensure that design-specific solutions are not imposed? This PART question evaluates whether the program requirements promote competition, modularity, and innovation by avoiding vendor-specific solutions. The sMOSA methodology promotes solution-independent requirements through product backlog development, user stories, modular decomposition, use of open standards, and supplier analyses. Requirements are iteratively refined and continuously validated through stakeholder feedback.

A8: To what extent do the system-level functional and performance specifications permit an open systems design? This PART question evaluates whether the system-level requirements and specifications maintain the implementation of the open systems architecture rather than relying on proprietary or vendor-specific solutions [1]. MOSA-compliant programs promote open systems design by designing for interoperability, compatibility, and open standards. These characteristics enhance the Agile Scrum cycle by incorporating the MOSA open-systems design into the sMOSA Product Roadmap Planning and the sMOSA Planning and Architect stages. Through the sMOSA methodology, architectural development, open standards, supplier analyses, and alternative assessments ensure the specifications maintain an open systems architecture [5]. In line with Agile Scrum practices, the specifications and requirements are reviewed, refined, verified, and updated as necessary with continuous stakeholder engagement.

A9: To what extent are modular, open system considerations included as part of alternative design analyses? This PART question evaluates whether modularity, open standards, and interoperability were considered when conducting trade studies, design solutions, and supplier selection [1]. Within the sMOSA methodology, identifying supplier options and assessing alternative configurations fall under the sMOSA Planning and Architect and Development and Integration stages; evaluating suppliers, modular decomposition, and standards maturity collectively inform the assessment of alternative solutions. The Agile Scrum practice encourages changing requirements and delivery in short IID cycles. As a result, the product increment can adapt to changing requirements and alternative designs. The MOSA pillars enhance the Agile Scrum cycle by including hardware within the software-centric design.

A10: To what extent are mechanisms established to migrate key interfaces that are proprietary or closed

to key interfaces that are open? This PART question evaluates whether the program has established a strategy to transition closed or proprietary interfaces and standards toward open status. Within the sMOSA methodology, standards evaluation begins with the Product Roadmap planning, including the MOSA Strategy Plan and the IP strategy. Through architectural development, allocating functions and interfaces to modular decomposition supports a design that relies less on proprietary implementations. Agile Scrum practices improve architectural openness through iterative, incremental design changes that incorporate open standards and interfaces.

A11: To what extent are MOSA principles reflected in the program's performance measures? This PART question evaluates whether the program established measurable metrics to assess the effectiveness of MOSA principles throughout the system lifecycle. The planning activities establish the performance measures. MOSA performance is measured continuously and reported to stakeholders to show progress and ensure the design and resulting prototypes adhere to MOSA principles. Deliverables are demonstrated, compatibility evaluated, and integration testing performed to ensure interfaces and standards are compatible. The Increment Release and Transition stage provides working MOSA deliverables, standardized interface specifications, and lessons learned as measurable indicators of MOSA implementation.

2. Technical Questions

B1: To what extent is the system's architecture based on related industry or other standard reference models and architectural frameworks? This PART question evaluates whether the program used architectural frameworks, a reference architecture, and industry standards acceptable within the industry, rather than developing proprietary architecture. During the sMOSA Product Roadmap Planning and sMOSA Planning and Architect stage, the team researches existing frameworks, architectures, and reference models, and surveys supplier options. The sMOSA process addresses this question through iterative architecture refinement and stakeholder feedback.

B2: To what extent is an architectural description language used to define system modules and interfaces? During the sMOSA Product Roadmap Planning and sMOSA Planning and Architect stages, the program is set up with common modeling languages, such as UML, SysML, DoDAF, UAF to communicate effectively, using such tools with outside developers. Model-Based Systems Engineering (MBSE) helps communicate and verify module boundaries, interface relationships, dependencies, and behaviors consistent with MOSA principles. "Model-Based Systems Engineering (MBSE) is a critical process which supports the activities required to achieve MOSA objectives" [6]. The sMOSA process supports these activities through iterative architectural refinement, requirements maturity, interface allocation, and stakeholder feedback. The resulting models become authoritative sources of truth (ASoT).

B3: To what extent does the system's architecture exhibit modular design characteristics? This PART question evaluates whether the system's architecture was intentionally designed using modularity characteristics of encapsulation, high cohesion, self-contained, and loosely-coupled that support independence, interchangeability, maintainability, scalability, and future technology upgrades [1]. Within the sMOSA process, in the sMOSA Planning and Architect stage, the system is intentionally decomposed into modular elements, the architecture is further developed, and singular functions are identified [5]. During the Development and Integration stage, the team refines the architecture and defines interfaces to create highly cohesive, loosely coupled modules. The team verifies and tests the modules for MOSA characteristics in the sMOSA Review and Retrospective stage. After the IID cycle, the team develops and integrates modules incrementally, maintaining modularity and keeping the system adaptable to changing requirements and future upgrades. Agile Scrum practices reinforce these activities through iterative backlog refinement, stakeholder review, and retrospective evaluation, allowing the team to assess and refine module boundaries and interfaces as the system evolves. Agile Scrum practices do not independently ensure modularity, but they provide the iterative planning, implementation, and continuous inspection (including stakeholder feedback) through which the MOSA design practices are repeatedly performed.

B4: To what extent is the system's architecture capable of adapting to evolving requirements and leveraging new technologies? This PART question evaluates the degree to which the system's architecture responds to change, encourages technology insertion to mitigate obsolescence, and supports longevity [1]. The sMOSA methodology and the IID cycle allow for changing requirements, updates to the last iteration, and product increments without major redesign. In the sMOSA Product Roadmap Planning stage, the Product Backlog and the updated Release Plan for the next iteration capture new or modified requirements and information debt. Within the sMOSA Planning and Architecture stage, modular decomposition and interface identification support changing requirements and future technological advancements. The Development and Integration stage supports changing requirements through backlog refinement, architectural updates, and the assessment of alternative configurations. The IID cycle allows stakeholders to provide

feedback on working increments. Lessons learned and captured information debt support modernization and continuous architectural improvement. Agile Scrum practices support this capability through continuous backlog refinement, iterative architecture development, stakeholder feedback, and the recurring evaluation of alternative configurations. When combined with modular decomposition, open standards, and well-defined interfaces in the sMOSA framework, these iterative practices support technology insertion while limiting redesign of unaffected modules.

B5: To what extent has the criteria for designating key interfaces been established? This PART question evaluates whether a specific methodology was established to identify key interfaces, as planned in the sMOSA Product Roadmap Planning stage, in conjunction with the MOSA strategy plan and architecture development. Key interfaces are evaluated based on their criticality, ease of integration, frequency of change, interoperability, and commonality [1]. In addition, the system architecture is refined, standards are assessed, and suppliers are evaluated to determine which interfaces are critical for interoperability, modularity, competition, and future technology insertion. Agile Scrum supports this development through stakeholder engagement, iterative reviews, and refinement of the architecture and interfaces.

B6: To what extent has the program designated key interfaces? Programs identify interfaces that are critical to interoperability, integration, sustainment, and modernization. Identifying key interfaces is a key pillar of MOSA. The modules interface and exchange information with host systems through external interfaces. The architecture identifies and controls the interfaces to ensure interoperability, sustainment, competition, and future upgrades. The architecture refines the interfaces further during decomposition and version-controls them throughout the stages. The key interfaces are verified and validated, and results are reviewed for unnecessary functions and interfaces that would hinder the module's modularity. The Increment Release and Transition stage captures the Interface Standardization Specifications to support modernization, reuse, and technology insertion. The Agile Scrum IID cycle allows changes and updates to the internal structure of the modules without impacting the key interfaces.

B7: To what extent has the program assessed the feasibility of using open standards for key interfaces? This PART question objectively evaluates whether standards exist and if the program conducted market, economic, technological, and operational studies that support the use of open standards for the identified key interfaces [1]. The standards feasibility assessment begins during the sMOSA Product Roadmap Planning stage through the MOSA Strategy Plan. The product scope and MOSA Strategy Plan establish the objectives and scope of the market analysis for interoperability, modularity, and standards use. During the sMOSA Planning and Architecture Refinement stage, teams evaluate standards for maturity and industry use through supplier analysis. The sMOSA process enhances this by iteratively evaluating and validating standards through prototype development. The methodology lets teams implement standards incrementally, demonstrating compatibility to stakeholders through working prototypes and integration testing, reducing the risk and uncertainty of adopting the down-selected standards.

B8: To what extent have standards selection criteria been established that give preference to open interface standards? This PART question evaluates the decision-making process and criteria for selecting and prioritizing open, consensus-based, and widely supported interface standards. The standards selection criteria are established during the sMOSA Product Roadmap Planning and sMOSA Planning and Architect stages and are governed by the MOSA Strategy plan, project, and product scope. During this stage, the team identifies and evaluates decision criteria for maturity, adoption, supplier availability, and lifecycle suitability.

B9: To what extent are open standards selected for key interfaces? While B7 evaluates the feasibility of using open standards, B8 evaluates the criteria for selecting standards, and B9 evaluates the outcome of those activities. Within the sMOSA methodology, the team identifies open standards, evaluates their maturity, and incorporates them into the system architecture. Supplier analysis and analysis of alternatives validate that the selected standards support competition and future technology insertion. Consistent with Agile Scrum principles, the team continuously validates standard selection through iterative development, integration, and stakeholder demonstrations, ensuring the architecture remains adaptable.

B10: To what extent are validation and verification mechanisms established to assure that system components and selected commercial products conform to the selected interface standards? The PART question evaluates whether the program has established formal verification and validation processes, policies, and testing activities to ensure that system components, including COTS, comply with selected open standards and interfaces. The team establishes V&V and testing strategies early in the sMOSA Product Roadmap Planning through the MOSA Strategy, Test, and Verification plans. The team continuously captures lessons learned and information debt during verification and testing activities. The team updates the corresponding plans based on lessons learned from the previous testing and verification iteration. During the Implementation/Development stage, continuous iterative and regression testing and verification activities identify incompatibilities and standards compliance issues, emphasizing the Agile Scrum

practice of continuous quality assurance and rapid feedback. Consistent with Agile Scrum practices, testing activities are continuous and flow into the sMOSA review and retrospective stage, where testing and verification increase to stakeholder demonstrations using flight, flight-like, or prototype hardware and software within Hardware-in-the-Loop environments.

B11: To what extent do system components and selected commercial products conform to standards selected for system interfaces? This PART question evaluates whether the intended MOSA components or COTS products have demonstrated actual conformance to the standards selected for the key interfaces. The B10 question asks whether V&V procedures, methods, and policies exist; B11 focuses on the results of V&V activities without requiring non-standard or proprietary adaptations or modifications. Continuous iterative and regression testing and verification activities are conducted during the Development and Integration stage to identify any incompatibilities and standards compliance issues, thereby emphasizing the Agile Scrum practice of continuous quality assurance and rapid feedback. Agile Scrum promotes continuous evaluation of working increments, allowing issues to be discovered early in the cycle. Consistent with Agile Scrum practices, testing activities are continuous and flow into the sMOSA review and retrospective stage, where testing and verification increase to include stakeholder demonstrations using flight, flight-like, or prototype hardware and software within Hardware-in-the-Loop environments.

B12: To what extent do system components and selected commercial products avoid utilization of vendor-unique extensions to interface standards? This PART question evaluates whether programs minimize the reliance on proprietary and vendor-specific interfaces, modifications, or extensions in the architecture or design. Vendor-specific interfaces create complexity in future enhancements, compatibility with other systems, and lifecycle longevity. During the sMOSA Product Roadmap Planning stage, guidelines and requirements for avoiding vendor-specific extensions are incorporated into the MOSA Strategy and Intellectual Property plans. During the sMOSA Planning and Architect stage, the team evaluates candidate standards and supplier analyses for maturity, industry acceptance, and support from multiple suppliers. During the Development and Integration stage, the team evaluates alternative configurations and allocates interfaces. Continuous testing and verification activities help identify proprietary dependencies. During the sMOSA Review and Retrospective, stakeholders demonstrate modules and conduct compatibility assessments to ensure they do not require vendor-specific modifications. Interfaces and standards are documented and controlled to support future modernization and acquisition activities. The sMOSA incorporates an IID approach through frequent integration, continuous testing, and stakeholder demonstrations and involvement. Testing the modules early and iteratively for compatibility and interoperability will identify proprietary design issues and technical debt before they present cost, schedule, or risk issues.

B13: To what extent can system components be substituted with similar components from competitive sources? This PART question evaluates whether the system architecture enables components and COTS products to be replaced with functionally equivalent alternatives from different suppliers without redesign. This activity begins with the sMOSA Planning and Architecture Refinement stage, which includes modular decomposition and identifying singular functions. These activities ensure highly cohesive, loosely coupled, and independent modules. Open standards and supplier analyses provide evidence of support from multiple suppliers. During the sMOSA Development and Integration stage, alternative configurations are considered to determine if different modules are compatible substitutes. Agile Scrum IDD supports design and testing iterations of alternatives to evaluate compatibility. The Increment Release and Transition stage supports future substitutes by standardizing the interface specifications and architectural knowledge for future upgrades.

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